

Labsheet-1

Real-World Data Collection, Cleaning and Preprocessing using Python

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In [3]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler

# 1. Load dataset
df = pd.read_csv("WA_Fn-UseC_-HR-Employee-Attrition-selected-columns.csv")

# 2. Inspect
print(df.head())
print("\nShape:", df.shape)
print("\nMissing Values:\n", df.isnull().sum())

# 3. Handle missing values
for col in df.select_dtypes(include=np.number):
    df[col] = df[col].fillna(df[col].median())

for col in df.select_dtypes(include="object"):
    df[col] = df[col].fillna(df[col].mode()[0])

# 4. Remove duplicates
df.drop_duplicates(inplace=True)

# 5. Clean categorical values
for col in df.select_dtypes(include="object"):
    df[col] = df[col].str.strip()

# 6. Feature Engineering
if "Age" in df.columns:
    df["AgeGroup"] = pd.cut(
        df["Age"],
        bins=[0, 25, 35, 45, 60, 100],
        labels=["Young", "Early", "Middle", "Senior", "Very Senior"]
    )

# 7. Outlier detection and treatment
for col in ["Age", "MonthlyIncome", "TotalWorkingYears"]:
    if col in df.columns:
        Q1 = df[col].quantile(0.25)
        Q3 = df[col].quantile(0.75)
        IQR = Q3 - Q1
        df[col] = df[col].clip(
            Q1 - 1.5 * IQR,
            Q3 + 1.5 * IQR
        )

# 8. Label Encoding
le = LabelEncoder()

for col in ["Attrition", "Gender", "OverTime"]:
    if col in df.columns:
        df[col] = le.fit_transform(df[col].astype(str))
```

```
# One-Hot Encoding
df = pd.get_dummies(df, drop_first=True)

# 9. Standardization
cols = [
    col for col in ["Age", "MonthlyIncome", "TotalWorkingYears"]
    if col in df.columns
]

if cols:
    df[cols] = StandardScaler().fit_transform(df[cols])

# 10. Save dataset
df.to_csv("IBM_HR_Preprocessed.csv", index=False)

# Final output
print("\nPreprocessing Completed!")
print("Final Shape:", df.shape)
print("Missing Values:", df.isnull().sum().sum())
print("\nFinal Data:")
print(df.head())
```

	Age	Attrition	BusinessTravel	DailyRate	Department \
0	41	Yes	Travel_Rarely	1102	Sales
1	49	No	Travel_Frequently	279	Research & Development
2	37	Yes	Travel_Rarely	1373	Research & Development
3	33	No	Travel_Frequently	1392	Research & Development
4	27	No	Travel_Rarely	591	Research & Development

	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber
0	1	2	Life Sciences	1	1
1	8	1	Life Sciences	1	2
2	2	2	Other	1	4
3	3	4	Life Sciences	1	5
4	2	1	Medical	1	7

Shape: (1470, 10)

Missing Values:

Age	0
Attrition	0
BusinessTravel	0
DailyRate	0
Department	0
DistanceFromHome	0
Education	0
EducationField	0
EmployeeCount	0
EmployeeNumber	0

dtype: int64

Preprocessing Completed!

Final Shape: (1470, 20)

Missing Values: 0

Final Data:

	Age	Attrition	DailyRate	DistanceFromHome	Education	EmployeeCount \
0	0.446350	1	1102	1	2	1
1	1.322365	0	279	8	1	1
2	0.008343	1	1373	2	2	1
3	-0.429664	0	1392	3	4	1
4	-1.086676	0	591	2	1	1

	EmployeeNumber	BusinessTravel_Travel_Frequently \
0	1	False
1	2	True
2	4	False
3	5	True
4	7	False

	BusinessTravel_Travel_Rarely	Department_Research & Development \
0	True	False
1	False	True
2	True	True
3	False	True
4	True	True

	Department_Sales	EducationField_Life Sciences	EducationField_Marketing \
0	True	True	False
1	False	True	False
2	False	False	False
3	False	True	False

4	False	False	False
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	EducationField_Medical	EducationField_Other	\
0	False	False	
1	False	False	
2	False	True	
3	False	False	
4	True	False	

	EducationField_Technical	Degree	AgeGroup_Early	AgeGroup_Middle	\
0		False	False	True	
1		False	False	False	
2		False	False	True	
3		False	True	False	
4		False	True	False	

	AgeGroup_Senior	AgeGroup_Very Senior
0	False	False
1	True	False
2	False	False
3	False	False
4	False	False